

ECO5050 AI Applications and Techniques in Economics

Lecture 9 Forecast Evaluation

Yaein Baek

Sogang University

Fall 2025

Outline

- 1 Pseudo Out-of-Sample Forecasts
- 2 Evaluation and Comparison of Multiple Forecasts

Pseudo Out-of-Sample Forecasts

- The many variants of out-of-sample (oos) forecast estimation methods differ in how they account for possible model instability by discounting past data more or less strongly.
- Example: A linear forecast model with a one-period forecast horizon $y_{t+1} = x_t' \beta + \varepsilon_{t+1}$ leads to one-step forecasts $\hat{f}_{t+1|t} = x_t' \hat{\beta}_t$, where

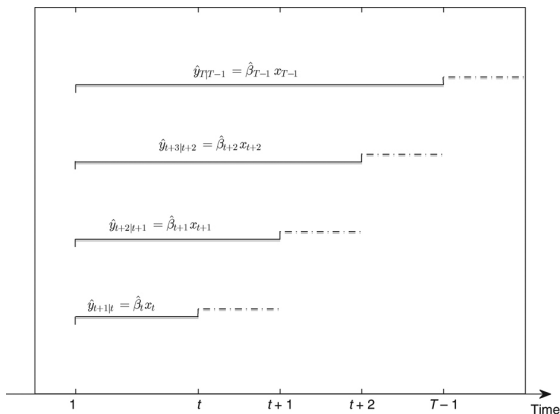
$$\hat{\beta}_t = \left(\sum_{s=1}^t w(s, t) x_{s-1} x_{s-1}' \right)^{-1} \left(\sum_{s=1}^t w(s, t) x_{s-1} y_s \right) \quad (1)$$

- Different methods are defined by different weight functions $w(s, t)$ which determine the importance of recent versus old data.
 - ▶ Expanding estimation window
 - ▶ Rolling estimation window
 - ▶ Fixed-proportion estimation window
 - ▶ Fixed estimation window
 - ▶ Exponentially declining weights

Pseudo Out-of-Sample Forecasts

Expanding estimation window

- Expanding or recursive estimation windows put equal weight on all observations from the beginning of the sample to the point of the forecast, $s = 1, \dots, t$, to estimate the parameters of the forecasting model.



Pseudo Out-of-Sample Forecasts

Expanding estimation window

- The expanding estimation window's weights in equation (1) take the form

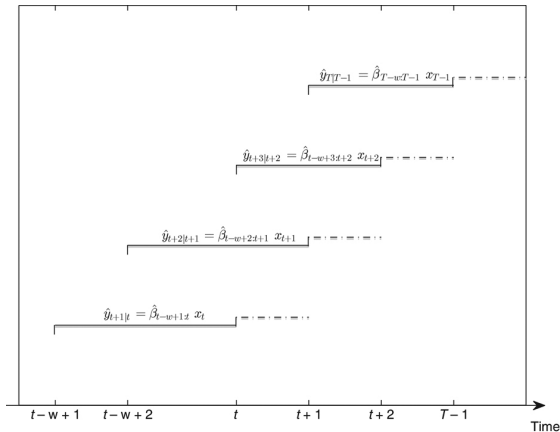
$$w(s, t) = \begin{cases} 1 & 1 \leq s \leq t, \\ 0 & \text{otherwise.} \end{cases}$$

- In a stationary world where the parameters of the data-generating process (DGP) does not change, the expanding window approach makes efficient use of the data and leads to consistent parameter estimates.
 - ▶ In a nonstationary world, the approach leads to biased forecasts.
- In empirical applications, producing relatively small parameter estimation errors often dominates the effect of reducing the squared bias, unless the change in the model parameters is very large (Pesaran and Timmermann, 2007)

Pseudo Out-of-Sample Forecasts

Rolling estimation window

- A rolling window uses an equal-weighted window of the most recent $\bar{w}$ observations to estimate the parameters of the forecasting window



Pseudo Out-of-Sample Forecasts

Rolling estimation window

- For example, to generate a forecast of y_{t+1} at time $t > \bar{w}$, data (y_s, x_{s-1}) , $s = t - \bar{w} + 1, \dots, t$ is used for parameter estimation. The rolling window weights in (1) is

$$w(s, t) = \begin{cases} 1 & t - \bar{w} + 1 \leq s \leq t, \\ 0 & \text{otherwise.} \end{cases}$$

- The rolling window method is typically used when forecasters are not convinced that the underlying DGP is stationary and so do not want to “contaminate” the model forecasts too much by basing them on old and potentially irrelevant data.

Pseudo Out-of-Sample Forecasts

Fixed-proportion estimation window

- Rather than using a fixed-length window, a fixed proportion λ of the data can be used to estimate the parameters of the forecasting model

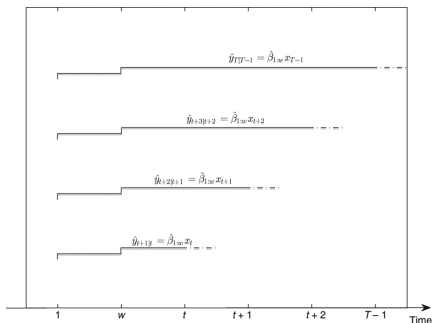
$$w(s, t) = \begin{cases} 1 & (1 - \lambda)t + 1 \leq s \leq t, \\ 0 & \text{otherwise.} \end{cases}$$

- This method is less widespread in applied work than the rolling (fixed) window method, but is sometimes used in analytical work to characterize the sampling distribution of various test statistics.

Pseudo Out-of-Sample Forecasts

Fixed estimation window

- The fixed window estimation method includes only the first $\bar{w}$ observations to once and for all estimate the parameters of the forecasting model used to generate all subsequent forecasts.



- This method is typically employed when the costs of estimation are very high and reestimating the model with new data is prohibitively expensive or impractical in real time.

Pseudo Out-of-Sample Forecasts

Exponentially declining weights

- In the presence of model instability, it is common to discount past observations using weights that get smaller, the older the data.

$$w(s, t) = \begin{cases} \lambda^{t-s} & 1 \leq s \leq t, \\ 0 & \text{otherwise.} \end{cases}$$

where $0 < \lambda < 1$

- Sometimes referred to as discounted least squares as the discount factor λ , puts less weight on past observations.

Inference of the Out-of-Sample Average Loss

- The pseudo out-of-sample approach seeks to mimic the real-time updating scheme underlying most forecasts.
- As the sample expands, the model parameters get updated, resulting in a sequence of forecasts. We can compute the average loss by comparing the realized values to these forecasts.
- West (1996) provides the main set of results that establishes conditions under which the estimated loss is both consistent and asymptotically normal under expanding, rolling, and fixed estimation windows.

Evaluation and Comparison of Multiple Forecasts

The Diebold-Mariano test

- The Diebold-Mariano (DM) test, proposed in Diebold and Mariano (1995), considers the forecast errors as primitives and makes assumptions directly on those errors.
 - ▶ “model-free” environment
- Suppose that two forecasts are available so we have two sets of losses, $L(f_{1t+h|t}, y_{t+h})$ and $L(f_{2t+h|t}, y_{t+h})$. Consider the difference

$$d_{t+h} = L(f_{2t+h|t}, y_{t+h}) - L(f_{1t+h|t}, y_{t+h})$$

- Test the null hypothesis

$$H_0 : E[d_{t+h}] = 0$$

- Diebold (2015) discuss the key assumptions: for all t

$$E[d_{t+h}] = \mu_d$$

$$\text{Cov}(d_t, d_{t-j}) = \gamma(j)$$

$$0 < \text{Var}(d_t) = \sigma^2 < \infty$$

Evaluation and Comparison of Multiple Forecasts

The Diebold-Mariano test

- The DM test statistic is given by

$$DM = \frac{\bar{d}_h}{\hat{\sigma}(\bar{d}_h)}$$

- ▶ $\bar{d}_h = T_P^{-1} \sum_{t=T_R+1}^{T-h} d_{t+h}$, $T_P = T - h - T_R$,
- ▶ $\hat{\sigma}(\bar{d}_h)$ is an estimator of the standard deviation of $\bar{d}_h$; Diebold and Mariano (1995) suggests estimating directly from the sequence $\{d_{t+h}\}$ using the method of Newey and West (1987)
- McCracken (2020) showed that if d_{t+h} is a covariance-stationary random variable with autocovariances decaying to zero,

$$DM \xrightarrow{d} N(0,1) \text{ as } T_P \rightarrow \infty$$

- We cannot conclude from the outcome of the Diebold-Mariano test which model is “best”; we can only make statements about which model generates the best forecast in a particular sample.

Evaluation and Comparison of Multiple Forecasts

The Diebold-Mariano test

- Harvey et al. (1997) modify the Diebold-Mariano test by making small sample adjustments

$$t_{MDM} = (1 + T_p^{-1}(1 - 2h) + T_p^{-2}h(h - 1))^{1/2}DM$$

where DM is the original Diebold-Mariano t -statistic.

- Suggests using the t distribution with $T_p - 1$ degrees of freedom, which is not based on theoretical concerns but serves to increase the critical values for tests that appear oversized in Monte Carlo experiments

Evaluation and Comparison of Multiple Forecasts

Finite sample predictability

- Test of population-level predictive ability: West (1996)

$$H_0 : E[L(f_{1t+1|t}(\beta_1^*), y_{t+1}) - L(f_{2t+1|t}(\beta_2^*), y_{t+1})] = 0 \quad (2)$$

Tests equality of forecasts at population values of the parameters, and the main goal is to validate population models

- Test of finite-sample predictive ability: Giacomini and White (2006)

$$H_0 : E[L(f_{1t+1|t}(\hat{\beta}_{1t}), y_{t+1}) - L(f_{2t+1|t}(\hat{\beta}_{2t}), y_{t+1})] = 0 \quad (3)$$

Tests equality of forecasts, given that the forecasts are constructed using estimated parameters $\hat{\beta}_{1t}$, $\hat{\beta}_{2t}$

Evaluation and Comparison of Multiple Forecasts

Finite sample predictability

- Giacomini and White (2006) tests (3), which retains the effect that estimation error have on forecasts.
 - ▶ Two models at time t are used to generate one-step-ahead forecasts, using a fixed window ω of observations.
 - ▶ Each forecast is a function of the data $(z_t, z_{t-1}, \dots, z_{t-\omega+1})$ and parameter estimates: $f_{it+1|t}(\hat{\beta}_{it})$, $i = 1, 2$
 - ▶ Construct losses $L(f_{it+1|t}(\hat{\beta}_{it}), y_{t+1})$ for $t = T_R + 1, \dots, T - 1$
- The effect of estimation error does not vanish in (3), which assumes a fixed estimation windows, whereas it vanishes in (2) which evaluates the expected losses at the probability limits of the estimators as the estimation sample gets very large.

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